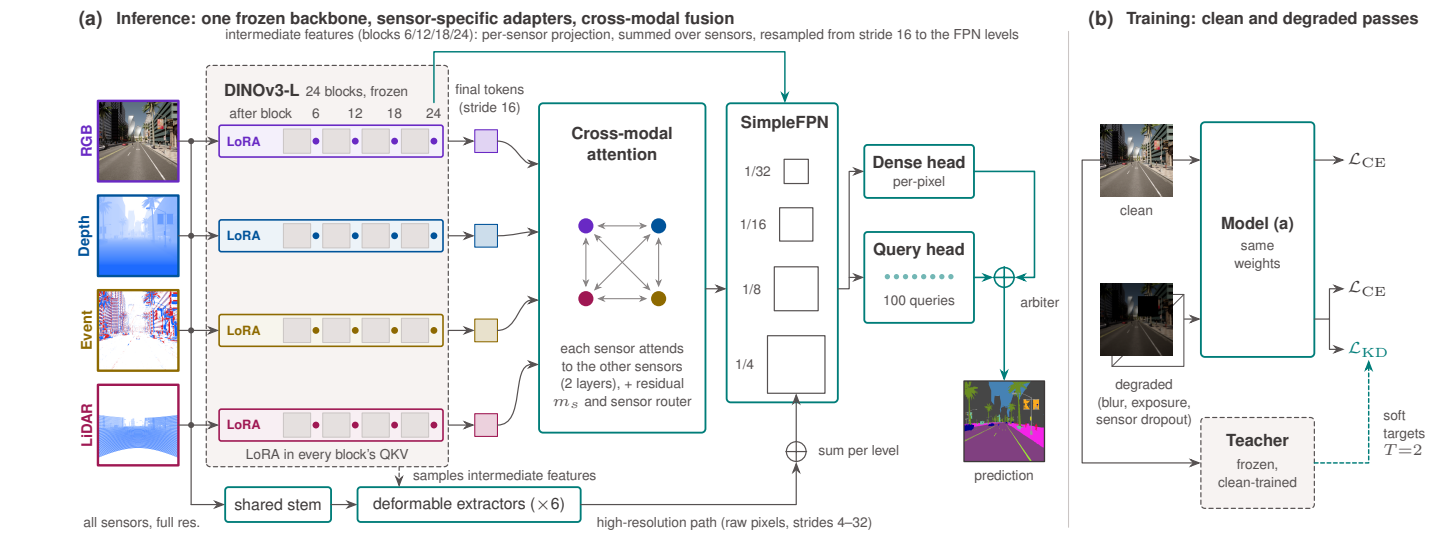

(a) Inference: one frozen backbone, sensor-specific adapters, cross-modal fusion
intermediate features (blocks 6/12/18/24): per-sensor projection, summed over sensors, resampled from stride 16 to the FPN levels
DINOv3-L 24 blocks, frozen
after block 6 12 18 24
final tokens (stride 16)
RGB
Depth
Event
LiDAR
LoRA
LoRA
LoRA
LoRA
LoRA in every block's QKV
Cross-modal attention
each sensor attends to the other sensors (2 layers), + residual m_s and sensor router
SimpleFPN
1/32
1/16
1/8
1/4
Dense head
per-pixel
Query head
100 queries
arbiter
prediction
sum per level
samples intermediate features
all sensors, full res.
shared stem
deformable extractors (×6)
high-resolution path (raw pixels, strides 4–32)
(b) Training: clean and degraded passes
clean
degraded (blur, exposure, sensor dropout)
Model (a)
same weights
$\mathcal{L}_{\mathrm{CE}}$
$\mathcal{L}_{\mathrm{CE}}$
$\mathcal{L}_{\mathrm{KD}}$
Teacher
frozen, clean-trained
soft targets $T=2$